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Mini Color NN

A Complete Color Quantization of CIFAR-10 Dataset for CNN Image Classification

Systems Research Yilmaz 2025

Introduction

Convolutional Neural Networks (CNNs) have revolutionized image classification but remain computationally intensive, thus limiting their deployment on low-resource devices such as surveillance cameras and IoT cameras. High dimensional image data requires significant memory and processing power, creating a bottleneck for real-time applications in more constrained environments. Mini-Color NN is aiming to reduce such computational demand.

Novelty

Unlike prior works focusing on grayscale datasets or low-bit quantization, this project applies Floyd-Steinberg Dithering combined with Vector Quantization (k-means clustering) to the entire full-color CIFAR-10 dataset. By propagating quantization error to neighboring pixels, the project preserves visual quality while compressing data. This is the first known effort to combine these two techniques a full-color benchmark dataset. The Mini-Color dataset is made publicly available under an open license, providing a new resource for research into lightweight neural network applications.

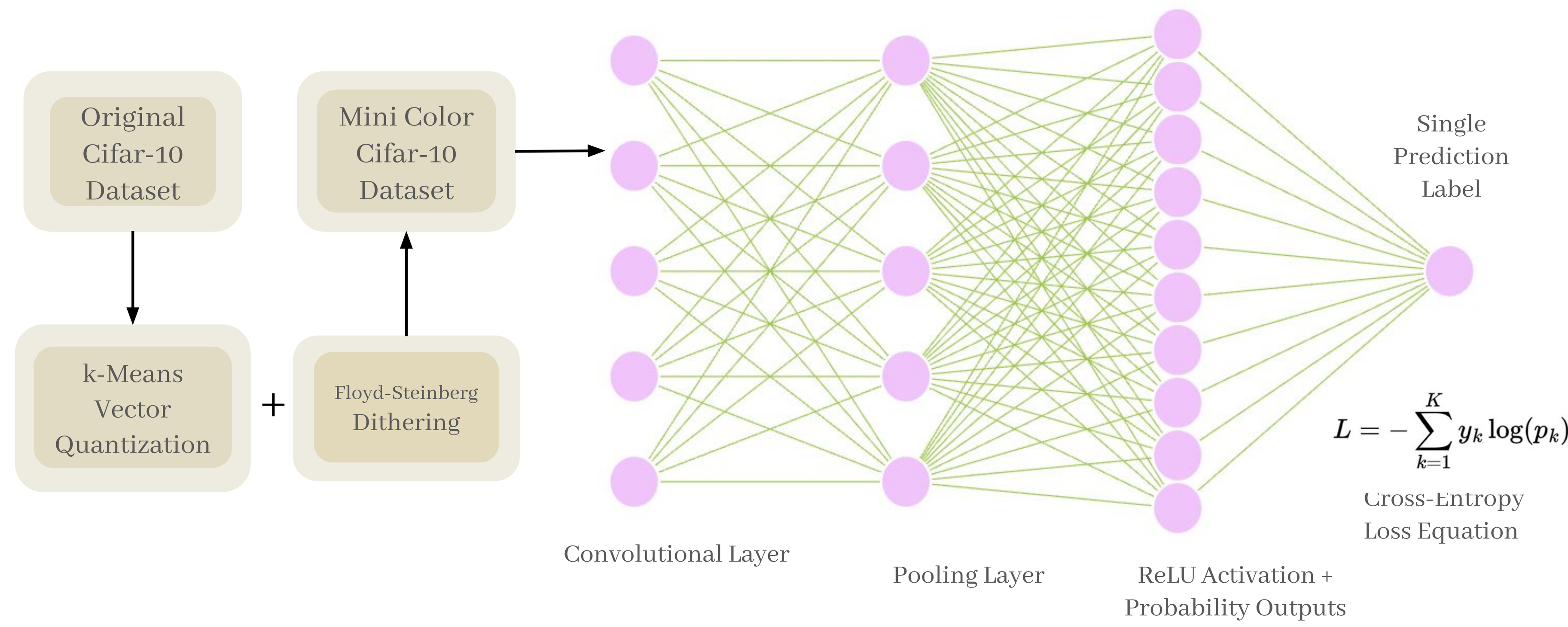

Figure 1: Systems Architecture

Methods

CIFAR-10 images were quantized using k-means clustering and then dithered pixel-by-pixel with Floyd-Steinberg error diffusion. As shown in *Figure. 1*, the new dataset was fed into CNNs, and models trained on both original and quantized dithered images were compared on testing accuracy, runtime, and memory efficiency. K values of 8 and 27 were used to compare the effect on cluster size, as shown in *Figure 2.*

Results

Using Mini-Color instead of original CIFAR-10 reduced training time by 1.3x and dataset size by 1.7x (80 mb reduction), with very comparable accuracy (92.88% vs 92.01%). Shown in Figure 3, loss increased slightly (0.0772 → 0.0927), but performance remained strong. This shows that the Mini-Color dataset offers efficient training and storage with minimal accuracy tradeoff.

Figure 2: Comparison of Original (L), Dithered 8 colors (M), and Dithered 27 colors (R)

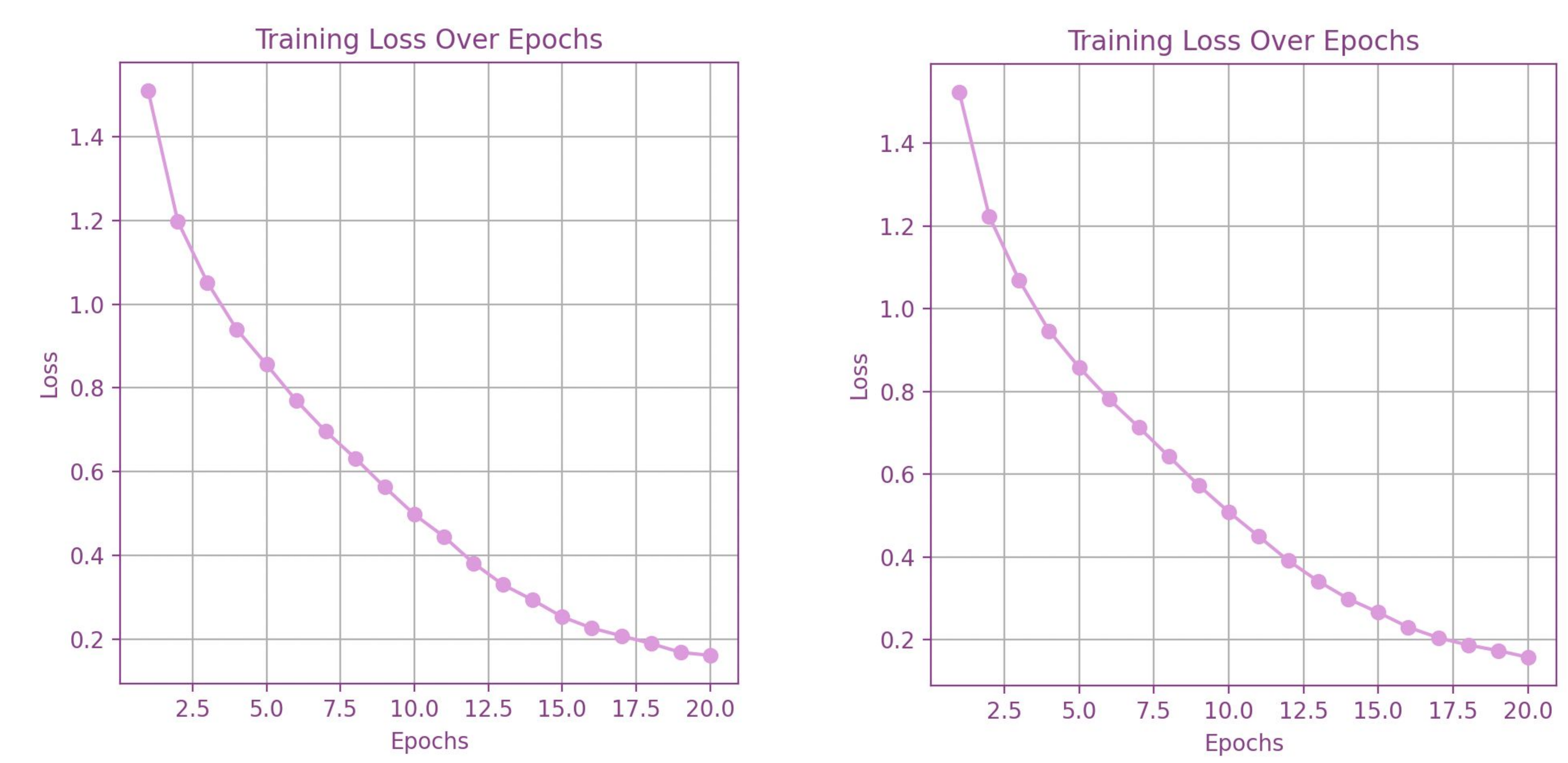

Figure 3: Side-by-side comparison of CNN Loss Graphs. Original (L) vs. Mini Color (R)

Impact & Future Work

Results highlight the potential of Mini-Color to enable faster, more storage-efficient neural networks with minimal loss in accuracy. Future work can explore improved quantization methods (e.g., K-means++), CNN tuning to further optimize performance, and real implementation.